Task 5b: Methods to compare against

Gap2Idea baseline experiment. Estimated effort: up to a week. Do methods 1-3 first.

Goal. Run the methods reviewers will expect beside our funnel, on the same test papers, and produce one table of methods by datasets with saved per-sentence predictions.

Methods

#	Method	Run on	Start from	Sources
1	**TF-IDF + logistic regression**; binary n-gram + BernoulliNB	SAL, FWS, arXiv. Saved totals exist for SAL and FWS, but rerun them to save predictions	`stage_c_package/baseline_grid.py`	Repo
2	**Kilicoglu rules**: section gate plus negative filters	SAL. It targets limitations in clinical papers. On arXiv, try only limitation labels using LaTeX sections, and report it as an extension. Not FWS	Java code: `RuleBasedLimitationSentenceRecognizer.java`. Sections: `section_name` in SAL `test.csv`; `data/gold/arxiv_latex_blocks.jsonl`	Kilicoglu et al. 2018 paper; github.com/kilicoglu/limitation-recognizer
3	**ModernBERT NLI alone**, no LLM	SAL, FWS. arXiv already scored: F1 0.614 on all gold test, needs arXiv-only split. Revision `ca476cb9`	`scripts/bench/nli_score_remote.py`, `nli_variants.py`, `stage_c_nli_qwen.py`	tasksource/ModernBERT-large-nli
4	**Fine-tuned SciBERT and BiomedBERT**	arXiv gold; FWS for SciBERT. SAL predictions are saved in `sal_test_allmodels.json` (PubMedBERT F1 0.604 after Stage A)	`scripts/training/stageb_v100.py`, `scripts/dataset/export_stageb_data.py`. Earlier runs failed because torch refused `.bin` weights; use safetensors or a newer torch	`allenai/scibert_scivocab_uncased`, `microsoft/BiomedNLP-BiomedBERT-base-uncased-abstract-fulltext` on Hugging Face
5a	**Lan et al. 2024**: fine-tuned PubMedBERT + EDA/PBA augmentation	SAL. Checkpoint not released, so reproduce it from the paper. Reported F1 0.821 uses a different pool (20.7% positives)	Ask Yazan for `h100_allmodels.py` (our PubMedBERT recipe)	Lan et al. 2024 paper; data: github.com/MengfeiLan/SAL_Type_Classification
5b	**Zhang et al.**: n-gram + chi-square + BernoulliNB	FWS. Reported macro-F1 0.907; our log says about 0.825 with chi-square inside folds. Rerun and save both	`scripts/bench/bench_fws.py`; `docs/experiments/fws_recognition.md`	Zhang et al. paper; data and code: github.com/xiangyi-njust/FWS

Data

Dataset	Train	Tune on	Test once	Get it from
SAL	`data/external/sal_rct/train.csv`	`dev.csv`	40 papers, 7,968 sentences: `sal_test_allmodels.json`. `test.csv` has 8,882 rows; 914 fragments are dropped, none positive	Private bundle, or github.com/MengfeiLan/SAL_Type_Classification
arXiv gold	`splits.json["train"]` (162 papers, keep arXiv for arXiv-only training)	`splits.json["val"]`	`load_val("test")`, `dom == "arxiv_cs"`: 28 papers, 10,388 sentences	Private bundle: `data/gold/gap_gold_final.jsonl`, `splits.json`, `data/paper_texts.jsonl`
FWS	Papers outside the test set in `fws_corpus.csv`	A paper-disjoint part of train	209 papers, 1,506 sentences: selection in `stage_c_package/breadth_funnel.py`, lines 67-75	Private bundle, or github.com/xiangyi-njust/FWS

Code: github.com/Yazangthb/Gap2Idea, branch `chore/repo-reorg`. GPU: ask Yazan for access. **Private bundle** = files gitignored in the repo. Ask Yazan for them. They contain labels, so anyone working with SAL or gold data cannot be a rater in the human study.

Steps

1. Check that the test papers match the counts above before training anything.
2. For each method, train on train and fix any threshold on the tuning split. Write the settings in `docs/experiments/baseline_methods.md` **before scoring test**.
3. Score test once by sentence: positive-class P, R, F1, with paper-level CI. Add macro-F1 for FWS, since published numbers use it.
4. Save predictions as JSONL with paper ID, sentence index and prediction, so paired tests with the funnel are possible. Paired CI code: `paired()` in `scripts/bench/stage_c_nli_qwen.py`.

Compare against

Dataset	Funnel	Earlier baselines (totals only, not comparable until rerun)
SAL-40	F1 0.673 (`stage_c_sal40_qwen.json`)	No Stage A: PubMedBERT 0.495, TF-IDF 0.443
arXiv gold	F1 0.641 (NLI + Qwen or bge-small + Qwen)	None on arXiv alone
FWS	F1 0.707 on Yandex; Qwen rerun from task 3	TF-IDF 0.714

Rules

- Never choose a threshold, prompt or model using test results. Report all methods tried, not only the best.
- Keep positive-class and macro-F1 apart; they are different numbers.
- If a published method cannot be reproduced, say why and give the closest result. Label published scores as "reported" when their pool differs from ours.

Done when

One table in `docs/experiments/baseline_methods.md` lists methods by datasets with P, R, F1 and CI, plus F1 difference from the funnel. Scripts and prediction files are saved, and anything not run is listed with the reason.